

Nonlinear Control System-Assignment1-Xinyi Yang

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Problem 1

Consider the time-varying linear system

$$\dot{x} = A(t)x.$$

(a) $A(t)$ is Hurwitz for each $t \geq 0$.

For a 2×2 real matrix, Hurwitz is equivalent to

$$\text{tr } A(t) < 0, \quad \det A(t) > 0.$$

Compute the trace:

$$\text{tr } A(t) = (-1 + 1.5 \cos^2 t) + (-1 + 1.5 \sin^2 t) = -0.5 < 0.$$

Compute the determinant. Let $s = \sin t$, $c = \cos t$.

$$a_{11}a_{22} = (-1 + 1.5c^2)(-1 + 1.5s^2) = -0.5 + 2.25s^2c^2.$$

$$a_{12}a_{21} = (1 - 1.5sc)(-1 - 1.5sc) = -1 + 2.25s^2c^2.$$

Hence

$$\det A(t) = (-0.5 + 2.25s^2c^2) - (-1 + 2.25s^2c^2) = 0.5 > 0.$$

Therefore $A(t)$ is Hurwitz for all $t \geq 0$.

(b) Verify that the proposed function is a solution.

Consider

$$x(t) = \begin{bmatrix} e^{0.5t} \cos t \\ -e^{0.5t} \sin t \end{bmatrix}.$$

Differentiate:

$$\dot{x} = e^{0.5t} \begin{bmatrix} 0.5 \cos t - \sin t \\ -0.5 \sin t - \cos t \end{bmatrix}.$$

Direct substitution shows that $\dot{x} = A(t)x$, hence the function is a solution of the system.

(c) Stability conclusion.

The solution contains the factor $e^{0.5t}$, hence

$$|x(t)| \rightarrow \infty.$$

Therefore the origin is not asymptotically stable.

This example shows that even if $A(t)$ is Hurwitz for every t , the linear time-varying system need not be stable.

Pointwise eigenvalues do not determine stability of LTV systems; uniform properties of the state transition matrix are required.

Problem 2

System:

$$\dot{x} = (6t \sin t - 2t)x.$$

The solution is

$$x(t) = x_0 \exp \left(\int_{t_0}^t (6\tau \sin \tau - 2\tau) d\tau \right).$$

Compute the integral:

$$\begin{aligned} \int 6\tau \sin \tau d\tau &= 6(\sin \tau - \tau \cos \tau), \\ \int -2\tau d\tau &= -\tau^2. \end{aligned}$$

Thus,

$$x(t) = x_0 \exp \left(6(\sin t - t \cos t - \sin t_0 + t_0 \cos t_0) - (t^2 - t_0^2) \right).$$

The exponent is dominated by $-t^2$ as $t \rightarrow \infty$, since $6 \sin t$ is bounded and $-6t \cos t = O(t)$. Hence the exponent tends to $-\infty$, so $x(t) \rightarrow 0$ for any initial condition.

Take $t_0 = 2n\pi$, so $\sin t_0 = 0$, $\cos t_0 = 1$. Then

$$x(t) = x_0 \exp \left(6(\sin t - t \cos t + 2n\pi) - (t^2 - 4n^2\pi^2) \right).$$

At $t = 2n\pi + \pi/2$, $\sin t = 1$, $\cos t = 0$, we get

$$x(t) = x_0 \exp \left(6(1 + 2n\pi) - ((2n\pi + \pi/2)^2 - 4n^2\pi^2) \right).$$

This grows with n , showing that for fixed t_0 , the solution can become arbitrarily large at a fixed time offset, so the origin is not uniformly stable.

Problem 3

System:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = -\sin x_1 - g(t)x_2, \quad g(t) \in [a, b], \quad 0 < a < b.$$

We use Matrosov's theorem to show uniform local asymptotic stability of the origin.

Choose the Lyapunov function

$$V(x) = 1 - \cos x_1 + \frac{1}{2}x_2^2.$$

Then

$$\dot{V} = \sin x_1 \cdot x_2 + x_2(-\sin x_1 - g(t)x_2) = -g(t)x_2^2 \leq -ax_2^2 \leq 0.$$

Thus $\dot{V} \leq 0$ and $\dot{V} \equiv 0$ implies $x_2 = 0$. From the system, $x_2 = 0$ gives $\dot{x}_2 = -\sin x_1 = 0$, so $x_1 = k\pi$.

To exclude nonzero equilibria, introduce an auxiliary function

$$W(x) = x_2.$$

Then

$$\dot{W} = -\sin x_1 - g(t)x_2.$$

On the set where $\dot{V} = 0$ (i.e., $x_2 = 0$), we have $\dot{W} = -\sin x_1$, which is nonzero for $x_1 \neq k\pi$. Hence Matrosov's theorem applies, and the origin is uniformly locally asymptotically stable.

Problem 4

System:

$$\dot{x} = (-1 + 2 \cos t) x.$$

We seek a Lyapunov function of the form

$$V(x, t) = \frac{1}{2} \exp(\alpha \sin t + \beta \cos t) x^2.$$

Let $p(t) = \alpha \sin t + \beta \cos t$. Then

$$\dot{V} = \frac{1}{2} \dot{p} e^p x^2 + e^p x \dot{x} = e^p x^2 \left(\frac{1}{2} \dot{p} - 1 + 2 \cos t \right).$$

We want the bracket to be negative and bounded away from zero. Choose $\alpha = -4$, $\beta = 0$, so $p(t) = -4 \sin t$, $\dot{p} = -4 \cos t$. Then

$$\frac{1}{2} \dot{p} - 1 + 2 \cos t = -2 \cos t - 1 + 2 \cos t = -1.$$

Thus

$$\dot{V} = -e^{-4 \sin t} x^2.$$

Since $e^{-4 \sin t} \geq e^{-4}$, we have

$$\dot{V} \leq -e^{-4} x^2.$$

Moreover,

$$V(x, t) = \frac{1}{2} e^{-4 \sin t} x^2 \geq \frac{1}{2} e^{-4} x^2.$$

Combining,

$$\dot{V} \leq -2e^{-4} \cdot \frac{1}{2} e^4 V = -2V.$$

Hence the origin is uniformly globally exponentially stable.

Problem 5

Since the origin of $\dot{x} = f(x)$ is GAS, there exists a smooth, positive definite, radially unbounded function $V_0(x)$ such that

$$\frac{\partial V_0}{\partial x} f(x) \leq -W(x) < 0, \quad \forall x \neq 0,$$

where $W(x)$ is positive definite.

For the time-varying system, consider the same $V(x) = V_0(x)$. Its derivative along trajectories is

$$\dot{V} = \frac{\partial V_0}{\partial x} g(t) f(x) = g(t) \frac{\partial V_0}{\partial x} f(x) \leq -g(t) W(x) \leq 0.$$

Thus $\dot{V} \leq 0$, and $\dot{V} \equiv 0$ implies $g(t) W(x) \equiv 0$. Since $g(\cdot)$ is PE, it cannot be identically zero on any interval; hence $W(x) \equiv 0$, which implies $x = 0$.

To apply Matrosov's theorem, we need an auxiliary function that is nonzero when $\dot{V} = 0$ but $x \neq 0$. Choose

$$V_1(x) = \frac{\partial V_0}{\partial x} f(x).$$

Then

$$\dot{V}_1 = \frac{d}{dt} \left(\frac{\partial V_0}{\partial x} f(x) \right) = \frac{\partial}{\partial x} \left(\frac{\partial V_0}{\partial x} f(x) \right) g(t) f(x).$$

On the set where $\dot{V} = 0$ (i.e., $g(t) = 0$ or $W(x) = 0$), we have $W(x) = 0 \Rightarrow x = 0$ by positive definiteness. If $g(t) = 0$ on an interval, PE prevents this except on a set of measure zero. Thus Matrosov's conditions are satisfied, and the origin is UGAS.

Problem 6

Consider the Lyapunov function candidate

$$V(x) = \frac{1}{2}x_1^2 + \frac{1}{2a}x_2^2.$$

Then

$$\dot{V} = x_1(-x_1 + x_2) + \frac{1}{a}x_2(ax_1 - ax_2 - g(t)x_2^3) = -x_1^2 + x_1x_2 + x_1x_2 - x_2^2 - \frac{g(t)}{a}x_2^4.$$

Simplify:

$$\dot{V} = -x_1^2 + 2x_1x_2 - x_2^2 - \frac{g(t)}{a}x_2^4 = -(x_1 - x_2)^2 - \frac{g(t)}{a}x_2^4 \leq 0.$$

Thus $\dot{V} \leq 0$, and $\dot{V} \equiv 0$ implies $x_1 = x_2$ and $g(t)x_2^4 = 0$. Since g is PE, it is not identically zero on any interval, so we must have $x_2 = 0$, hence $x_1 = 0$. Therefore the only invariant set where $\dot{V} = 0$ is the origin.

To apply Matrosov's theorem, introduce function

$$W(x) = x_1 - x_2.$$

Then

$$\dot{W} = \dot{x}_1 - \dot{x}_2 = (-x_1 + x_2) - (ax_1 - ax_2 - g(t)x_2^3) = -(1+a)(x_1 - x_2) + g(t)x_2^3.$$

On the set where $\dot{V} = 0$ (i.e., $x_1 = x_2$ and $g(t)x_2^4 = 0$), we have either $x_2 = 0$ or $g(t) = 0$. If $x_2 = 0$, then $x = 0$. If $g(t) = 0$ on an interval, PE prevents this. Hence Matrosov's conditions hold, and the origin is UGAS.

Problem 7

$$(6) \quad \dot{x}_1 = -x_1 - x_1^3 + x_2, \quad \dot{x}_2 = -x_1 - x_2 + u$$

Consider $V(x) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$. Then

$$\dot{V} = x_1(-x_1 - x_1^3 + x_2) + x_2(-x_1 - x_2 + u) = -x_1^2 - x_1^4 + x_1x_2 - x_1x_2 - x_2^2 + x_2u = -x_1^2 - x_1^4 - x_2^2 + x_2u.$$

Using Young's inequality $x_2u \leq \frac{1}{2}x_2^2 + \frac{1}{2}u^2$, we get

$$\dot{V} \leq -x_1^2 - x_1^4 - \frac{1}{2}x_2^2 + \frac{1}{2}u^2.$$

Thus $\dot{V} \leq -\frac{1}{2}\|x\|^2 + \frac{1}{2}u^2$ for $\|x\|$ large enough. Hence the system is ISS with linear gain.

$$(10) \quad \dot{x}_1 = -x_1 + x_2, \quad \dot{x}_2 = -h(x_1) - x_2 + u$$

Given $h(0) = 0$ and $yh(y) \geq ay^2$ for all y , with $a > 0$. Consider

$$V(x) = \int_0^{x_1} h(s) ds + \frac{1}{2}x_2^2.$$

Then

$$\dot{V} = h(x_1)(-x_1 + x_2) + x_2(-h(x_1) - x_2 + u) = -x_1h(x_1) + h(x_1)x_2 - x_2h(x_1) - x_2^2 + x_2u = -x_1h(x_1) - x_2^2 + x_2u.$$

Using $x_1h(x_1) \geq ax_1^2$ and Young's inequality $x_2u \leq \frac{1}{2}x_2^2 + \frac{1}{2}u^2$, we obtain

$$\dot{V} \leq -ax_1^2 - x_2^2 + \frac{1}{2}x_2^2 + \frac{1}{2}u^2 = -ax_1^2 - \frac{1}{2}x_2^2 + \frac{1}{2}u^2.$$

Thus $\dot{V} \leq -\min\{a, \frac{1}{2}\}\|x\|^2 + \frac{1}{2}u^2$, so the system is ISS with linear gain.

Problem 8

System:

$$\begin{aligned} \dot{x}_1 &= -x_1 + g(t)x_2, \\ \dot{x}_2 &= -g(t)x_1 + g(t)x_3, \\ \dot{x}_3 &= -g(t)x_2 + u, \end{aligned}$$

with $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ continuously differentiable, bounded, with bounded derivative, and $t \mapsto g^2(t)$ persistently exciting.

For $u = 0$, the system is $\dot{x} = A(t)x$ with

$$A(t) = \begin{bmatrix} -1 & g(t) & 0 \\ -g(t) & 0 & g(t) \\ 0 & -g(t) & 0 \end{bmatrix}.$$

Consider the Lyapunov function candidate $V_0(x) = \frac{1}{2}\|x\|^2$. Then

$$\dot{V}_0 = x^T A(t)x = -x_1^2 + g(t)x_1x_2 - g(t)x_2x_1 + g(t)x_2x_3 - g(t)x_3x_2 = -x_1^2.$$

Thus $\dot{V}_0 \leq 0$, and $\dot{V}_0 = 0$ implies $x_1 = 0$. Then the system reduces to

$$\dot{x}_2 = g(t)x_3, \quad \dot{x}_3 = -g(t)x_2.$$

Define $W(x) = x_2^2 + x_3^2$. Then $\dot{W} = 2g(t)x_2x_3 - 2g(t)x_3x_2 = 0$, so W is constant. With $x_1 = 0$, we need to show that the only solution that stays in $\{x_1 = 0\}$ is $x = 0$. Use an auxiliary function $W_1(x) = x_2$. Then $\dot{W}_1 = g(t)x_3$. On the set where $\dot{V}_0 = 0$ (i.e., $x_1 = 0$), if $x_3 \neq 0$ and $g(t)$ is PE, then $\dot{W}_1$ cannot be identically zero unless $x_3 = 0$. Similarly, $x_3 = 0$ then implies $x_2 = 0$. Hence Matrosov's theorem applies, and the origin of $\dot{x} = A(t)x$ is UGAS.

Since $A(t)$ is continuous, bounded, and the origin is UGAS, there exists a continuously differentiable matrix function $P(t)$ such that for all $t \geq 0$,

$$c_1 I \leq P(t) = P(t)^T \leq c_2 I,$$

and

$$\dot{P}(t) + A(t)^T P(t) + P(t) A(t) \leq -I.$$

Define $V(x, t) = x^T P(t)x$. Then along trajectories of the forced system $\dot{x} = A(t)x + Bu$ with $B = [0, 0, 1]^T$, we have

$$\dot{V} = x^T(\dot{P} + A^T P + P A)x + 2x^T P B u.$$

$$\dot{V} \leq -\|x\|^2 + 2\|PB\|\|x\|\|u\| \leq -\|x\|^2 + 2c_2\|x\|\|u\|.$$

Completing the square or using Young's inequality, for any $\theta > 0$,

$$2c_2\|x\|\|u\| \leq \theta\|x\|^2 + \frac{c_2^2}{\theta}u^2.$$

Choosing $\theta = 1/2$, we get

$$\dot{V} \leq -\frac{1}{2}\|x\|^2 + 2c_2^2u^2.$$

Since $V \geq c_1\|x\|^2$, we have $\|x\|^2 \leq V/c_1$, and thus

$$\dot{V} \leq -\frac{1}{2c_1}V + 2c_2^2u^2.$$

This differential inequality implies ISS with linear gain. More precisely, for any initial state and input, the solution satisfies

$$\|x(t)\| \leq \beta(\|x(0)\|, t) + \gamma\|u\|_\infty,$$

with $\gamma = \sqrt{4c_1c_2^2}$.